

logic timing diagrams from your simulation activity, which can then be exported to PDF, PNG and SVG formats.

Logisim

Logisim is an educational tool for designing and simulating digital logic circuits. With its simple toolbar interface and simulation of circuits as you build them, it is simple enough to facilitate learning of the most basic logic circuit concepts. With the capacity to build larger circuits from smaller sub-circuits, and to draw bundles of wires with a single mouse drag, Logisim can be used to design and simulate entire CPUs for educational purposes.

The software comes with a large base of documentation. The documentation not only includes the features of the software but also tutorials, examples and guides. Designed specifically for students, this software is very user-friendly.

OpenSCADA

openSCADA is an Open Source supervisory control and data acquisition system. It is platform-independent and based on a modern system design that provides security and flexibility at the same time. It is not an out-of-the box solution, but a set of tools that can be combined in many different ways. It provides development libraries, interface applications, mass configuration tools, and front-end and back-end applications.

SZARP

SZARP is a complete, full-featured, Open Source SCADA HMI software, available for free under GPL 2.0 conditions. It is designed for supervisory control of municipal heating systems.

The software has been actively developed for over 15 years and is currently used in over 40 heating plants all over Poland. A SZARP server can gather data from all kinds of industrial devices. There are over 30 device drivers available for this software at present.

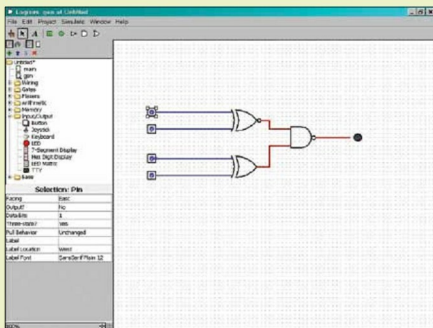


Fig. 2: Designing digital circuit in Logisim

SZARP contains built-in Lua scripting language for defining parameters based on actual or historical data values. Historical or actual data can be easily distributed over the Internet through SZARP.

Gentbvhdl

Gentbvhdl has been developed primarily for those who are learning VHDL and want to do a quick validation of their design. It can also be used for design purpose. One such application could be when the RTL owner needs to compile, elaborate and run the 0th time simulation. The 0th time simulation is required to ensure that the simulation bring-up is correct. There are certain design issues which do not show up until the design gets simulated at the 0th time. Download and extract the tar file and follow the simple steps/commands to generate the test bench.

FSMDesigner

FSMDesigner is a C++ based implementation for a finite-state machine (FSM) design tool with integrated hardware description language (HDL) generation. FSMDesigner4 uses the Simple-Moore FSM model guaranteeing efficient, fast, complex

control circuits. It can create finite-state machines for hardware projects with a comfortable GUI. Users can export their FSM to Verilog HDL or VHDL. The software also features export of FSMs for project documentation.

IndigoSCADA

IndigoSCADA is a small-footprint SCADA system entirely developed in 'C' and C++ with multiple OS support and multiple front-end protocol drivers. It includes features such as use of data to generate daily/weekly/monthly/management reports, historical and real-time graphical presentation of data, real-time event and alarm notification, and integrated real-time and historical databases. It also has an SQL editor to allow on-line maintenance of real-time and historical databases. It also has support for multiple users with different access rights, easy backups and client server architecture for fast database transactions. DDS architecture with the software enables real-time inter-process communication of SCADA processes.

Additionally, this software includes facility for soft logic programming with 'C' scripts with built-in IEC